Strictly Positive Parity Laws and the Limits of Marginal Certificates

Exact Heat-Bath Spectrum and Optimal Variance Tensorization

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Abstract

On the Boolean cube, the strictly positive laws with density $1 + t\prod_{i=1}^{n} x_i$, $|t| < 1$, have identical proper coordinate marginals. We compute their complete single-site heat-bath spectrum for arbitrary positive site rates. If R is the sum of the rates and m their minimum, the gap is $[R - \sqrt{(R-2m)^2 + 4t^2 m(R-m)}]/2$. Its inverse is the exact optimal variance tensorization constant. This constant diverges as $|t| \to 1$ at fixed dimension and rates, although all proper marginal data stay fixed. Thus no finite-valued function of those data bounds the constant over all strictly positive laws. This quantitative obstruction requires neither a hard parity constraint nor a comparison functional vanishing at the product law. We also state the older exact-zero certificate argument with a literal input map and distinguish genuine full-conditional coefficients from marginal proxies. The construction is classical and the spectral calculation is elementary; priority of the particular formula is not established. No fixed-local, spatial-decay, entropy-constant or physical mass-gap theorem is claimed.

1 The law and the meaning of marginal input

Let $n \geq 2$, $\Omega = \{-1, 1\}^n$, and let u be its uniform product law. Write $[n] = \{1, \dots, n\}$, $\chi_S(x) = \prod_{i \in S} x_i$ and $S^c = [n] \setminus S$. Define

$$\mu_t(x) = 2^{-n}(1 + t\chi_{[n]}(x)), \qquad -1 < t < 1. \tag{1}$$

Normalization follows from $\mathbb{E}_u \chi_{[n]} = 0$, and every configuration has strictly positive probability. Integrating over any omitted coordinate removes the perturbation, so every proper marginal equals the corresponding uniform product marginal. This is the classical parity mechanism behind limited-order independence; no novelty is claimed for it [1].

For a law μ, define the literal input tuple

$$M_r(\mu) = (\mu_S : S \subseteq [n],\ |S| \leq r),$$

where coordinate labels are retained. Saying that a functional of (μ, f) has this marginal input means an actual factorization $B(\mu, f) = \Psi(M_r(\mu), f)$, with the same fixed observable f when laws are compared. No conditional quantity is admitted merely because it conditions on few variables. In fact, from (1),

$$\mu_t(X_i = s \mid x_j,\ j \neq i) = \frac{1 + ts\chi_{[n]\setminus\{i\}}(x)}{2}. \tag{2}$$

These conditionals vary with t while $M_{n-1}(\mu_t) = M_{n-1}(u)$.

2 Complete spectrum with arbitrary site rates

Fix rates $\rho_i > 0$, independently of t, and put $R = \sum_i \rho_i$, $m = \min_i \rho_i$. Let $P_i f = \mathbb{E}_{\mu_t}[f \mid x_j,\ j \neq i]$ and use the positive operator

$$H = \sum_i \rho_i (I - P_i). \tag{3}$$

Thus the Markov generator is $-H$. Each site is refreshed at rate ρ_i; we do not divide by R. The usual full-conditional heat-bath and variance tensorization conventions are recorded, for unit rates, in [2, eqs. (1.3), (1.5), (1.6)].

Theorem 1. *The constant function has eigenvalue zero and $\chi_{[n]} - t$ has eigenvalue R. For each unordered pair $\{S, S^c\}$ with S nonempty and proper, put $a = \sum_{i \in S} \rho_i$ and $b = R - a$. The invariant space* $\operatorname{span}(\chi_S, \chi_{S^c})$ *has eigenvalues*

$$\lambda_\pm(a) = \frac{R \pm \sqrt{(a-b)^2 + 4t^2 ab}}{2}. \tag{4}$$

These spaces and the two distinguished modes exhaust the spectrum, with repetitions counted by blocks. In particular,

$$\operatorname{gap}_\rho(t) = \frac{R - \sqrt{(R-2m)^2 + 4t^2 m(R-m)}}{2} > 0. \tag{5}$$

Proof. Equation (2) gives $P_i \chi_S = \chi_S$ when $i \notin S$ and $P_i \chi_S = t \chi_{S^c}$ when $i \in S$. Therefore

$$H\chi_S = a\chi_S - ta\chi_{S^c}. \tag{6}$$

For the empty set and full set this proves the distinguished modes. For a proper nonempty pair, the coefficient matrix in the displayed ordered basis is

$$B(a) = \begin{pmatrix} a & -tb \\ -ta & b \end{pmatrix}. \tag{7}$$

Its trace is R and determinant is $(1-t^2)ab$, giving (4); both roots are positive. The off-diagonal rates must not be interchanged when $a \neq b$.

For completeness and orthogonality, character multiplication gives

$$\mathbb{E}_{\mu_t}[\chi_S \chi_T] = \mathbf{1}_{S=T} + t\mathbf{1}_{S=T^c}. \tag{8}$$

Different complement pairs are orthogonal in $L^2(\mu_t)$. Each pair has positive definite Gram matrix $\left(\begin{smallmatrix} 1 & t \\ t & 1 \end{smallmatrix}\right)$. The 2^n characters form a basis, consisting of the empty/full pair and $2^{n-1} - 1$ proper pairs. Each P_i is an orthogonal projection, so H is self-adjoint; these blocks account for all eigenvalues.

For every proper nonempty S, both a and $R - a$ are at least m. Hence $a(R-a) \geq m(R-m)$, with equality for a singleton of minimum rate. The lower root $[R - \sqrt{R^2 - 4(1-t^2)a(R-a)}]/2$ increases with $a(R-a)$. It is minimized at that singleton, is at most $R/2$, and is no larger than any upper root or the separate eigenvalue R. This proves (5) and its strict positivity. □

Corollary 2. *The optimal constant in*

$$\operatorname{Var}_{\mu_t}(f) \leq C_\rho(t) \sum_i \rho_i \mathbb{E}_{\mu_t}[\operatorname{Var}_{\mu_t}(f \mid x_j,\ j \neq i)] \tag{9}$$

for all real f is $C_\rho(t) = 1/\operatorname{gap}_\rho(t)$.

Proof. Orthogonal projection gives the exact Dirichlet form identity

$$\langle f, Hf\rangle_{\mu_t} = \sum_i \rho_i \mathbb{E}_{\mu_t}[\mathrm{Var}_{\mu_t}(f \mid x_j,\ j \neq i)].$$

The finite-dimensional spectral decomposition on the mean-zero space therefore proves the inequality and its sharpness. More explicitly, for $t \neq 0$, a minimum-rate singleton S and $\lambda = \mathrm{gap}_\rho(t)$, an extremizer is

$$f = \chi_S + \frac{m-\lambda}{t(R-m)}\chi_{S^c}.$$

Substitution in (7) proves the eigenvector equation. Its mean is zero and variance is positive by (8). At $t = 0$, χ_S alone is an extremizer. □

Remark 3 (Limits and normalization)**.** *Rationalizing* (5) *yields*

$$\mathrm{gap}_\rho(t) = \frac{2(1-t^2)m(R-m)}{R + \sqrt{(R-2m)^2 + 4t^2 m(R-m)}}. \tag{10}$$

Consequently $(1-t^2)C_\rho(t) \to R/[m(R-m)]$ *as* $|t| \to 1$*, at fixed dimension and rates. For unit rates the gap is* $[n - \sqrt{(n-2)^2 + 4(n-1)t^2}]/2$*; it equals* $1 - |t|$ *at* $n = 2$ *and equals* 1 *at* $t = 0$*. At fixed* $|t| < 1$ *it tends to* $1 - t^2$ *as* $n \to \infty$*. The random-scan transition operator* $P = R^{-1}\sum_i \rho_i P_i$ *has gap* $\mathrm{gap}_\rho(t)/R$*. None of these limit statements is inferred from finite diagnostics.*

3 A quantitative obstruction without exact-zero certification

For an arbitrary strictly positive law μ on the cube, write $C_\rho(\mu)$ for the optimal constant in (9) with its actual conditionals.

Theorem 4. *Fix* $n \geq 2$ *and the positive site rates. There is no finite-valued function* F *of all proper coordinate marginals such that*

$$C_\rho(\mu) \leq F(M_{n-1}(\mu))$$

for every strictly positive law μ *on the cube. Equivalently, no strictly positive gap lower bound depending only on this tuple can hold on that whole class.*

Proof. All μ_t in (1) have the same tuple M_{n-1}. A finite F at this common input is a fixed number, while Theorem 1 and (10) show $C_\rho(\mu_t) \to \infty$ as $|t| \to 1$. Every member of this sequence is strictly positive. The positive-gap formulation is the reciprocal of the same contradiction. □

Taking $n = r + 1$ proves the obstruction for each fixed marginal order $r \geq 1$. Neither a product-law right-hand side equal to zero nor a law with zero-probability configurations is used in this theorem. This does not exclude certificates on a class with a specified uniform probability lower bound, bounded interaction parameters or other substantive structural information. Here the ratio of largest to smallest configuration probability is $(1 + |t|)/(1 - |t|)$, which is unbounded along the obstructing sequence. Also μ_t is proportional to $\exp(\mathrm{atanh}(t)\chi_{[n]})$: the interaction is global, not a fixed local pair potential.

4 The restricted exact-zero lemma and its scope

The earlier elementary certificate argument remains useful if its input and zero condition are stated literally.

Proposition 5. *Let $B(\mu, f) = \Psi(M_r(\mu), f)$ and fix a single observable f. If laws μ, ν have identical M_r, if $B(\nu, f) = 0$, and if $D(\mu, f) > 0$, then $D(\mu, f) \le cB(\mu, f)$ fails for every finite c. A claimed measure-class application must establish all these conditions on actual formulas.*

Proof. Factorization gives $B(\mu, f) = B(\nu, f) = 0$, contradicting the asserted positive left-hand side. □

For example, at unit rates, take $f = X_n$ and $D(\mu, f) = \mathrm{Var}_\mu(f) - \sum_i \mathbb{E}_\mu \mathrm{Var}_\mu(f \mid \mathrm{rest})$. Under u this defect is zero. Under μ_t, $\mathrm{Var}(f) = 1$, only the $i = n$ conditional term is nonzero, and that term is $1 - t^2$ by (2). Thus $D(\mu_t, f) = t^2$. The exact-zero lemma still assumes its displayed certification condition; Theorem 4 is a different quantitative statement that does not assume it.

True full-conditional coefficients do not have the marginal-input factorization above. For the usual total-variation influence, flipping x_j changes the conditional Bernoulli bias at site $i \neq j$, giving $c_{ij} = |t|$ and row sum $(n-1)|t|$. These values vary while all proper marginals stay fixed. No unspecified Dobrushin-to-entropy constant is invoked. The entropy theorem in [2, Theorem 2.1] has its own ergodicity and conditional-coefficient hypotheses; it is not contradicted by the present construction.

For $n \ge 3$, $\mathrm{Cov}_{\mu_t}(X_i, X_j) = 0$ when $i \neq j$. Ordinary pair-coordinate covariance therefore cannot separate these laws. The specified block observables X_n and $\prod_{i<n} X_i$ have covariance t, but no spatial-decay conclusion follows without a geometry and allowed support dependence of constants. Classical Efron–Stein bounds under independence are not refuted. The previous claimed application to four audited inequalities is withdrawn here: no four-formula bridge establishing factorization, certification and separation has been supplied.

5 Evidence, comparison and limits

The complete proof is analytical. A finite exact rational diagnostic constructs the actual conditional resampling matrix, rather than assuming its Fourier block. It checks its action on all characters, detailed balance, Gram self-adjointness, block dimension counts, trace, determinant and the minimum subset product. The cases cover $n = 2, \ldots, 6$, $t = 0, 3/5, -9/10$, and both unit and unequal rational site rates: thirty cases and 32736 state-character identities in each mode. On an owned Colab Linux CPU runtime, normal and optimized Python each exited zero and produced byte-identical complete reports. These finite cases check implementation consistency; they do not replace the all-n proof. No new Lean/Lake proof or formal replay of the spectral theorem is claimed.

Immutable proof, diagnostic, reports and source bindings are available at [3]; the scientific diagnostic source is `3b5160dc7b7cbaea8a797fa36f0babf0fe9b8b91`. The historical Lean character-sum artifact belongs to the older note; it is not a formalization of the new spectrum or optimal constant.

The parity family and limited-marginal independence are classical. The calculation is an explicit specialization for a global interaction. A targeted primary-source search did not establish priority; absence of a located match is not novelty evidence. No entropy constant, fixed-local Gibbs, spatial-decay, gauge-field, continuum or physical mass-gap theorem is claimed.

Model assessments and review context

Codex assisted the analytic derivation, source preparation and diagnostic orchestration. The exact served backend and reasoning effort are not exposed in this preparation context. Prior manuscript participation by the preparing agent is explicit; independence is not asserted. This revised PDF has not yet received its declared exact-artifact scientific assessment. Any later numerical assessment must identify these exact PDF bytes and retain the complete original report, objections and factual model context. No historical grade is transferred to this version. AI assessment, formal checking, deposit, editorial acceptance and human peer review are distinct.

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